



FREGATA UAV

Material Selection, Empennage Preliminary Design and Pylon Design



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Abstract

On average every year more than 3000 migrants die trying to cross the Mediterranean Sea from Libya in an attempt to reach the Italian coast. The Fregata UAV designed by FLR22 is a response to this issue that seeks to minimize future loss of life around large bodies of water. This report goes through the Material Selection Process, Empennage Design and Pylon Design of a 51.5 kg MTOW, VTOL-fixed wing, electric-combustion hybrid, UAV capable of operation from boat decks. The empennage preliminary design follows a minimalist approach to structural design, ignoring traditionally used aircraft structural components such as stringers and multiple ribs for the sake of minimizing weight. Similarly, the pylon design was optimized to respond to geometrical constraints while still fulfilling its role to support the VTOL motors. The material selection process will be carried out to select materials that minimize weight and cost and are adequate for operation in marine environments. The step-by-step design process included numerous trade-offs and compromises both in the design itself but also in relation to the requirements of other subgroups.

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1 Introduction & Literature Review

1.1 Baseline and Objective

The aircraft that was conceptualized by the FLR22 group to best respond to the evolving crisis in the Mediterranean Sea was a 51.5 kg, payload dropping, VTOL, fixed wing, UAV. The baseline configuration of the Fregata UAV was a high-wing configuration with constant quarter-chord, and a H-tail empennage. It was decided to use 4 electric VTOL engines mounted on pylons emanating from the wing, with an combustion engine powered pusher-prop aft of the fuselage. The undercarriage was chosen as a deployable airbag mechanism [5]. In an effort to address the market of the Fregata UAV, dimensional constraints were imposed to limit the span and length of the UAV to 2.5 m and 2 m respectively, so as to facilitate its launch from boat decks. The material selection and design of the structural components was done with minimal weight and cost in mind with the goal of maximizing useful payload and housing as many useful and difference-making systems (like thermal cameras) as possible on-board.

1.2 Initial Weight Sizing

It was important to perform a conceptual weight estimation for the different structural components to obtain initial inertia loading scenarios for the first iteration of the preliminary design. This first guess leads to an initial preliminary sizing and a new, more accurate value for weight. The new weight value will then be used to size the structural components again. This iterative process helps to gradually improve the accuracy of the weight estimates for the components. Traditionally, these estimates would be made during the conceptual design of the aircraft using empirical methods based on commercial aircraft data as exemplified by Raymers' Aircraft Conceptual Design Book [6]. However given the lack of clear historical equivalents to the aircraft type being designed it would've been inaccurate to use existing databases for general aviation aircrafts and related equations for the initial weight guesses of the empennage of our light, VTOL, UAV. Therefore a dual method was undertaken to attempt to reach a reasonable first guess for the empennage. A thesis was found that attempts to adapt existing sizing equations of aircrafts to TUAVs using modern commercial, fixed-wing UAV data, by modifying the constants used in such equations [7]. The results from this method were then compared with a simpler method relying purely on geometry to estimate the volume of the empennage with the initial sizing dimensions. Both values converged enough that a value could be taken as an average of the two methods. This resulted in an initial weight guess for the HTP and VTP of 2 kg and 0.75 kg respectively.

2 Material Selection

The material selection process can be highlighted by the following chart, which starts with the market need to be addressed and the proposed solution. From there the selection process for materials to cut down from the continuously expanding available materials to find one that is the right fit for the project. This is done firstly by translating the design requirements, screening against identified constraints, before ranking materials using objective based functions or merit indexes and performing the final selection with any additional criterias that did not make it into the overall selection process, like the ready availability of data and/or documentation for the potential materials.

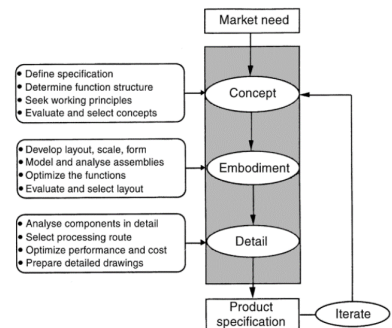


Figure 2.1: Material Selection Process [1]

2.1 Operating Conditions and Constraints

Identifying the operating conditions of the aircraft and other potential constraints that result directly from its intended uses is needed to perform an initial screening of materials. Materials whose physical properties cannot handle or operate within the constraints set by the design and its operating environment should be filtered out as soon as possible.

The first step was to identify the service and general operating conditions of the aircraft. The core design principle of the presented UAV is that it will operate over and/or around large bodies of water, in particular in the harsh atmospheric conditions of the open sea, more specifically in the Mediterranean. Therefore, a general idea of the service conditions was obtained using meteorological data from an island called 'Lampedusa' [2] that is located between Libya and Italy where the most migrant crossings are performed a year. This process identified peak surface temperatures of around 40°C, with minimums of around 10°C. Safety factors were taken into account, similarly to the design of a general aviation aircraft, to establish a minimum service operating temperature of -20°C and maximum expected service temperature of 50°C. Materials unable to operate below 50°C and above -20°C were filtered out.

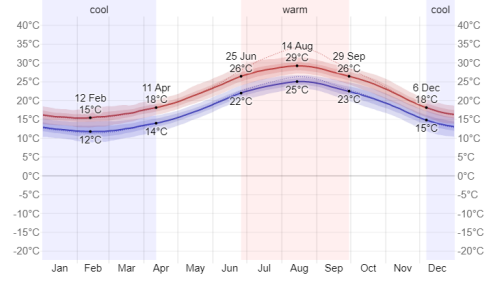


Figure 2.2: Yearly T° in Lampedusa [2]

Another crucial aspect revealed by this analysis was the rate of salinity in the air, as well as the peak humidity. With a salinity proportion of 36 ppt in the air [8], this puts the region slightly above the global average of 34 ppt, while peak humidity reached was, at times of the year, 98% [2]. These factors clearly indicated that the selected materials would need to be adequate for use in sea water and marine environments, which was an added constraint. These constraints were used for every single component discussed below.

2.2 Monocoque Skin

The reasoning behind the decision to use a monocoque skin revolved around the minimal design approach decided in order to save weight. The benefits in terms of weight of a monocoque design are due to the fact that it relies on the skin to resist and redistribute the loads the aircraft is subjected to. This implies that the loads applied on the aircraft must not cause the skin to buckle, the limitation of monocoque design, thereby removing the need and added mass of stringers and light frames [9]. Given the expected loads on the aircraft were small, it was assumed that a monocoque design would be mostly feasible for the structure, if only for the inclusion of a few heavy frames to resist point loads. Finally, a monocoque design was deemed to be beneficial due to the implication that the entirety of the aircraft outer skin would be manufactured from the same material. This serves to reduce the rapid galvanic corrosion caused when two different metals interact in salt water environments [10]. Another layer of safety against this phenomenon was taken by coating the aircraft with protective paint, as explored by Borsu Lefevre [11].

2.3 Optimization and Merit Index

The methodology for material selection followed from the earlier definition of the operating constraints, and the assumptions of a monocoque skin. The main components for which material selection was identified as a priority were the canopy, the skin of the monocoque, the different load carrying structures of the wing, and fuselage heavy frames, as well as the pylons. Table 2.1 establishes the different loading scenarios expected for each part of the structure and the corresponding index against which the materials that respond to the constraints will be ranked. Since the design seeks both minimal weight and cost, every index is for weight or cost optimized design [1].

Table 2.1: Material selection drivers and Merit Index

Component	Loading	Driver	Merit Index
Canopy	Impact	Damage	$K_{IC}/E\rho C_m$
Skin	Comp/Ten., Torsion, Buckling	Strength	$\sigma_C^{1/2}/\rho C_m$
Spars/Ribs	Comp/Ten., Bending, Shear, Buckling	Stiffness/ Strength	$E^{1/2}/\rho C_m$ or $\sigma_Y^{2/3}/\rho C_m$
Heavy Frames	Comp/Ten., Bending, Shear, Torsion	Strength	$\sigma_Y^{2/3}/\rho C_m$
Pylons	Bending, Comp/Ten.	Stiffness/ Strength	$E^{1/2}/\rho$ or $\sigma_C^{1/2}/\rho$

2.4 Load Carrying Structures

Using the Merit Indexes laid out in table 2.1, the CES Edupack Software package [3] was used to plot Ashby Charts for the different load carrying structural components, namely the skin, spars/ribs in the wing, and the heavy frames of the fuselage. One of the aims of the design was to have homogeneous materials for all these structures so as to reduce the risk of galvanic corrosion and decrease costs as the material for manufacture could be purchased in bulk. As figures 2.3a and 2.3b highlight for both the stiffness and strength limited designs, Aluminium 7055-T74511 was a material which a high index performance for all the structures. Other recurring materials were stainless steel, martensitic, as well as coated steel and die cast Aluminium 518.0. These were discarded on the basis of their lower performance for price to density ratio, shown in figure A.1 in the Appendix, as well as due to their worse structural integrity post)manufacturing (for die cast Aluminium [12]). Aluminium 7055-T74511 was therefore selected.

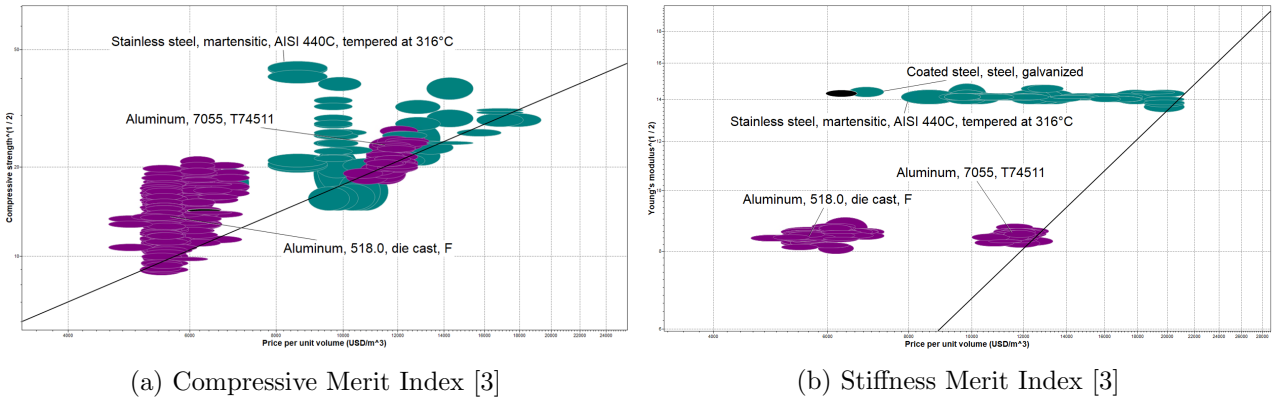


Figure 2.3: Stiffness and Strength Considerations

2.5 Pylons

Due to their particular position and loading, detailed below in section 4, a material was sought that yielded high compressive strengths and bending stiffnesses while also minimizing weight. As figure 2.4 demonstrates, a high performing material in that regard was an epoxy/HS carbon fiber composite. It was selected over other highlighted options like BMI/HS carbon fiber due to the more extensive and available documentation on the composite.

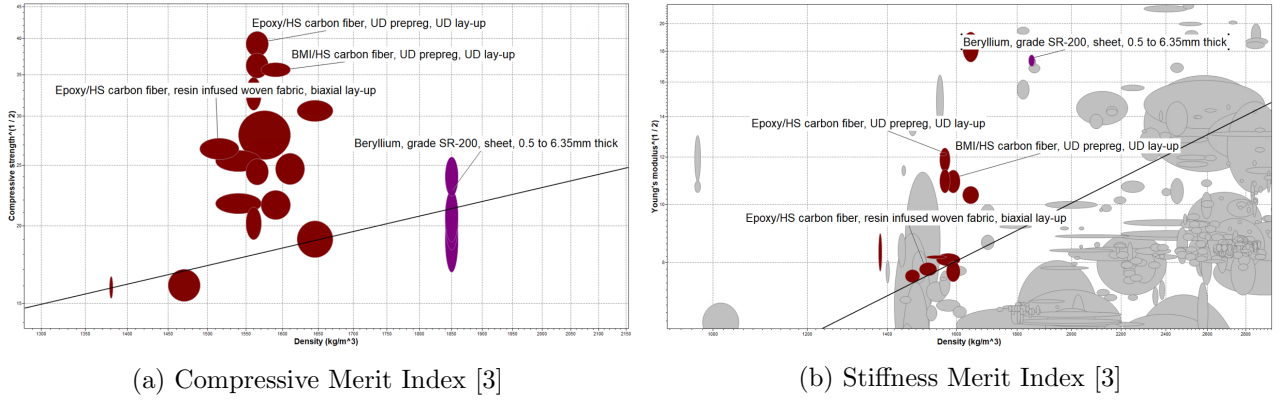


Figure 2.4: Stiffness and Strength Considerations for the Pylon

2.6 Canopy

2.6.1 Canopy Skin

The canopy is of particular importance as it is the part of the fuselage that shields the costliest equipment onboard. The combination of the avionic systems, estimated at around £7000, and the search and rescue systems with a thermal camera, estimated at around £5500 represents 68% of the total UAV cost. It was therefore decided that it would be made of the same material as the rest of the fuselage, given its high strength and impact resistance properties, and so as to fit into the monocoque design concept.

2.6.2 Canopy - Camera Cut-Out

The decision to house the thermal camera inside the canopy was made to reduce outer attachments to the fuselage to lower drag and avoid possible damage to such expensive equipment. This implies having a small part of the canopy that allows for both Long Wave Infrared (LWIR - 8-14 μ m [4]) and visible light (0.4-0.8 μ m) to pass since a regular camera was also included. The material chosen for this role was zinc sulfide (ZnS), also known as Cleartran [13], given its large transparency range (0.35-14 μ m) as seen in figure A.3 and good mechanical properties for materials of this type ($E = 74.5$ GPa).

2.7 Final Material Selection

The final results of the material selection for each component has been gathered into table 2.2 below:

Table 2.2: Final Material Selection [3]

Component	Material	ρ (kg/m ³)	E (GPa)	Price (£/kg)	σ_Y (MPa)	σ_C (MPa)	σ_T (MPa)	K_{IC}
Canopy	Cleartran (ZnS)	4090	74.5	-	-	60	60	1
Skin	Aluminium 7055 T74511	2850	72.7	4.29	581.5	567.5	627	36.15
Spars/Ribs	Aluminium 7055 T74511	2850	72.7	4.29	581.5	567.5	627	36.15
Frames	Aluminium 7055 T74511	2850	72.7	4.29	581.5	567.5	627	36.15
Pylons	Epoxy/HS CF, resin infused woven fabric, biaxial layup	1515	61	19	653.5	710	653.5	31.2

3 Empennage

Given the nature of the aircraft and the dimensional constraints created by the varying propellers around the fuselage, it was decided that the most beneficial empennage configuration that would clash least with the other components of the aircraft was a H-tail. The structural design of the empennage was therefore done for a regular H-tail, where the horizontal tail is attached to the fuselage at its centre, and the vertical tail is split into two components, each on one end of a half-horizontal tail [6].

The structural design of the tailplane was done in response to the sizings made by the aerodynamics [14] and control [15] subgroups that optimized the VTP and HTP lifting surfaces while ensuring their ability to create sufficient moments to ensure lateral stability and pitching authority. The design aimed to minimize the weight of the empennage while also ensuring it could support the required loading.

3.1 VTP

3.1.1 Loads

The loads on the vertical tail originate from two sources, inertia and lift. The inertia load, in this case the 'own-weight' of the VTP, was distributed along the tail as various point loads and it will contribute to both the shear and torque experienced by the structure. It will not contribute to the bending moments however, given the use of a symmetrical NACA 0010 airfoil. The main other load on the VTP is the lift needed to counter the maximum maneuver force identified by the V-N diagram from Itthithavorn [16]. With a load factor of $n = 3$ for this maximum load case, it was determined that the maximum lift required by the VTP was 1515.6 N. This force should then be divided by two since it will be distributed to the two vertical tail components of the H-tail. This loading yields the bending moment, shear force and torque distributions pictured in Figures 3.1a, 3.1b & 3.1c below:

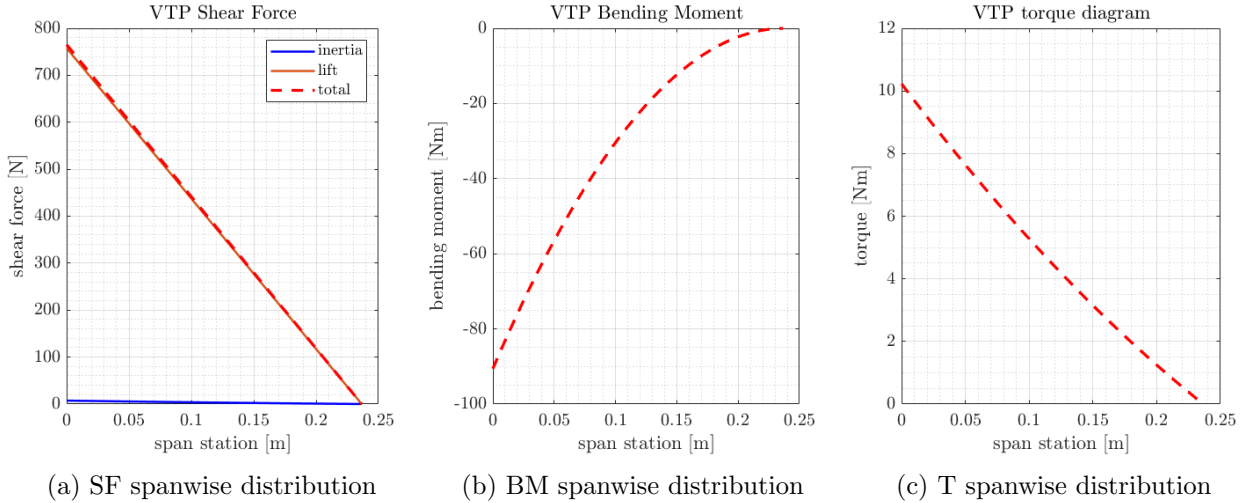


Figure 3.1: Loads on individual VTP

3.1.2 Design

The design of the vertical tail plane was done following conventional aircraft lifting surface design [17]. A traditional 2 spar design was selected with the front spar located at 10% of the chord, and the rear spar at 60% of the chord. The front spar location was selected so that it could connect into the front spar of the HTP underneath, also at 10% of the chord, ensuring a good redistribution of the inertial loads in exchange for a minimal increase in torque on the structure. The rear spar location, meanwhile, was selected to support the location of the rudder hinge and carry the hinge moments

expected at that location [15]. Given the overall small nature of the forces on the tail, it was decided that a simple 2 rib design would suffice to maintain the aerodynamic shape of the airfoil and resist possible crush loads. These were simply placed at the root and tip of the span. Stringers were not considered to improve the efficiency of the structure since the loads were minimal. The final geometry of the structure can be observed in figure 3.2.

Having decided on the geometry of the VTP, the widths (chords), and heights of the wing box (wb) (shaded on figure 3.2) could be obtained. It was then necessary to calculate the thicknesses of the different components for the combined loading from shear force, bending moments and torque.

3.1.3 Skin

The skin of the vertical tailplane was designed to resist compressive and tensile loads, and checked for torsion and buckling. From the load cases considered above, the bending moment was first converted to a compressive force per unit width per equation 1 from which the thickness of the skin could be obtained for the different span locations using equation 2 below:

$$N = \frac{BM}{(0.45c)(0.10c)} + \frac{SF}{c_{wb}} \quad (1)$$

$$t_{skin} = \sqrt[3]{\frac{N(c_{wb})^2}{3.62E_{skin}}} \quad (2)$$

where c_{wb} is taken as the length between the two spars at each span location of the wing box. The nominal stress in the skin could then be obtained as $\sigma_{buckling} = N/t_{skin}$.

3.1.4 Ribs

The ribs are designed to resist crush loads across the surface and increase the torsion and buckling resistance of the skin. As per the geometry that was decided, they are positioned at the root and the tip of the VTP, separated by a $b_{rib} = span = 0.1673m$. Part of their role is to increase the skin panel second moment of area, I_x that will resist crush loads. Therefore the crush load (in Newton) can be calculated from equation 3, and then be used to calculate the optimal rib thickness per equation 4:

$$F_{crush} = \frac{(BM)^2 b_{rib} h_{wb} t_{skin} c_{wb}}{2EI_x^2} \quad (3)$$

$$t_{rib} = \sqrt[3]{\frac{F(h_{wb})^2}{3.62E_{skin}c_{wb}}} \quad (4)$$

3.1.5 Spars

The spars of the vertical tail are designed to resist the combined loading from bending, shear and torsion. By combining the shear flow from the estimated shear forces and torque separately as $q_{SF} = SF/(2h_{wb})$ and $q_T = T/(2h_{wb}c_{wb})$ respectively. While the shear flow from shear force is assumed to be equally split between the front and rear spar, the torque shear flow is unevenly split since for a counter-clockwise bending moment it increases the flow around the front spar but decreases it for the rear spar. Combining with a buckling coefficient K_s obtained from ESDU datasheets D.1 for conservative design, the front and rear spar thicknesses could be obtained from equations 5 & 6 below:

$$t_{FS} = \sqrt[3]{\frac{(q_{SF} + q_T)h_{wb}}{K_s E}} \quad (5)$$

$$t_{RS} = \sqrt[3]{\frac{(q_{SF} - q_T)h_{wb}}{K_s E}} \quad (6)$$

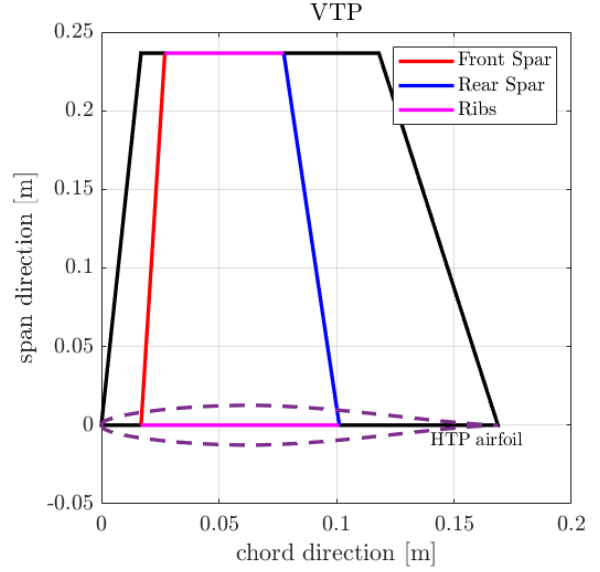


Figure 3.2: VTP Wing Box Geometry

3.1.6 Results

The design procedure detailed above yielded the following results for the corresponding skin, rib and spar thicknesses, where a limiting minimum manufacturing thickness of 1 mm has been chosen, as per industry standards mentioned in AVD lectures [17]. Where the required thicknesses are below this limit, they will be set to 1 mm.

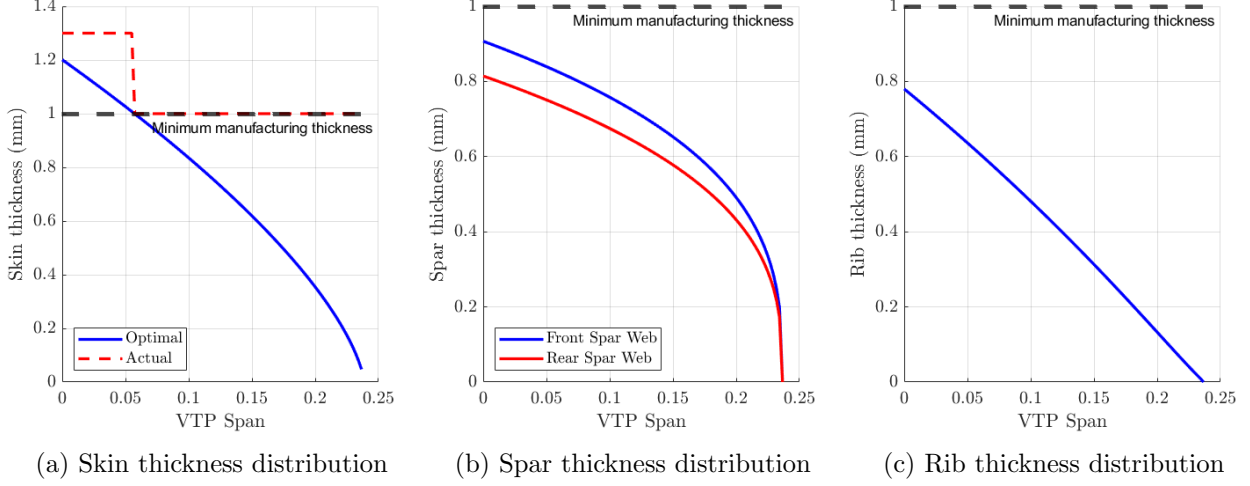


Figure 3.3: VTP Component Thicknesses

From these values, a buckling analysis can be performed for the skin panels. Using the actual skin thickness from figure 3.3a, the compressive stress ratio R_c and shear stress ratio R_s due to torque shear flow $\tau_s = q_T/t_{skin}$ in the skin can be calculated and the worse case scenario combined loading $R_{cs} = R_s^2 + R_c$ obtained, yielding a maximum value of 0.67, far below the limit of 0.99.

$$R_c = \frac{\sigma_{buckling}}{3.62E_{skin}(t_{skin}/c_{wb})^2} \quad (7)$$

$$R_s = \frac{\tau_s}{\sigma_{yield_{Al7055-T7}}} \quad (8)$$

3.2 HTP

3.2.1 Loads

The design of the HTP differs little from that of the VTP, if not for the fact that the loads are distributed slightly differently. This is in part due to the different geometry reflected in figure 3.4.

For inertia loads the main difference comes from the H-tail design and the fact that the vertical tailplanes are located at the end of the span. This weight was distributed across the last few span locations of the HTP where the VTP overlaps. Additionally, given the use of the NACA 64015 airfoil, the inertia loads will contribute to bending moments. The required lift force required by the HTP to pull the aircraft from a dive was estimated at 2215 N for a load factor of 4.5 [15]. This loading yields the bending moment, shear force and torque distributions pictured in Figures 3.5a, 3.5b & 3.5c.

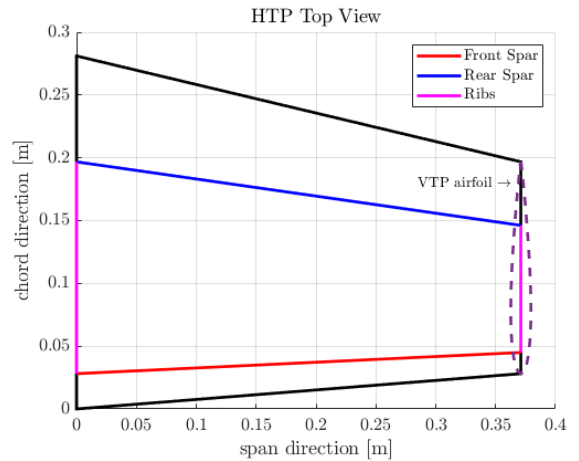


Figure 3.4: HTP Wing Box Geometry

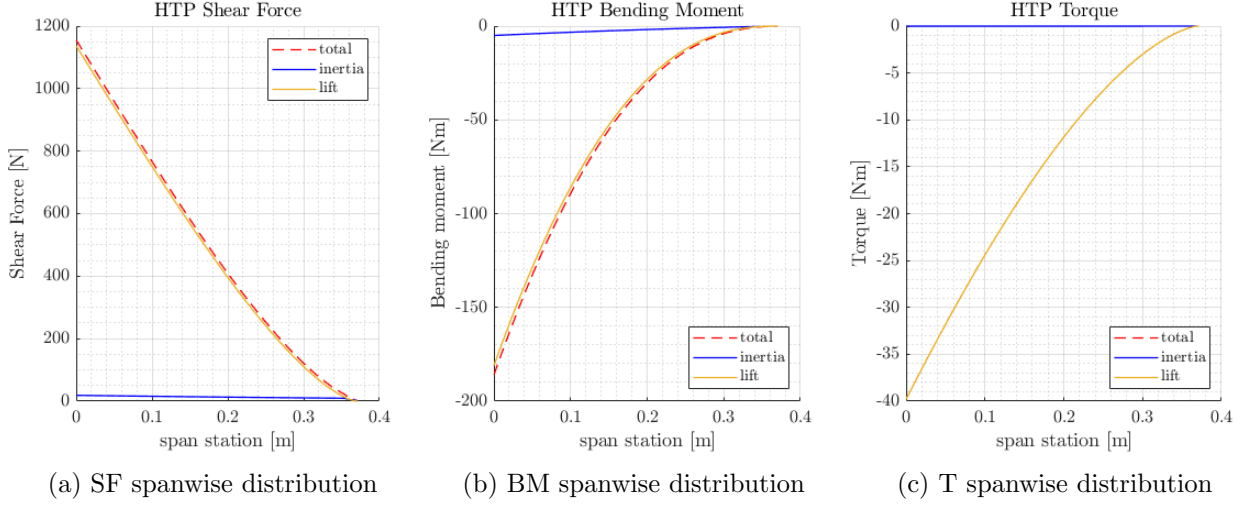


Figure 3.5: Loads on HTP half-span

3.2.2 Skin, Spars and Ribs

The design of the skin, spars and ribs of the HTP followed the same methodology as for the VTP. The main difference is with regards to the torque resultant on the front and rear spars. Given the direction of lift on the HTP is downwards, the resulting torque is in the opposite direction to that of the VTP. This results in an opposite torque shear flow distribution to the spars from equations 5 & 6, and therefore the rear spar is expected to be thicker. The required thickness distribution for the HTP skin, spars and ribs can be found below in figure 3.6:

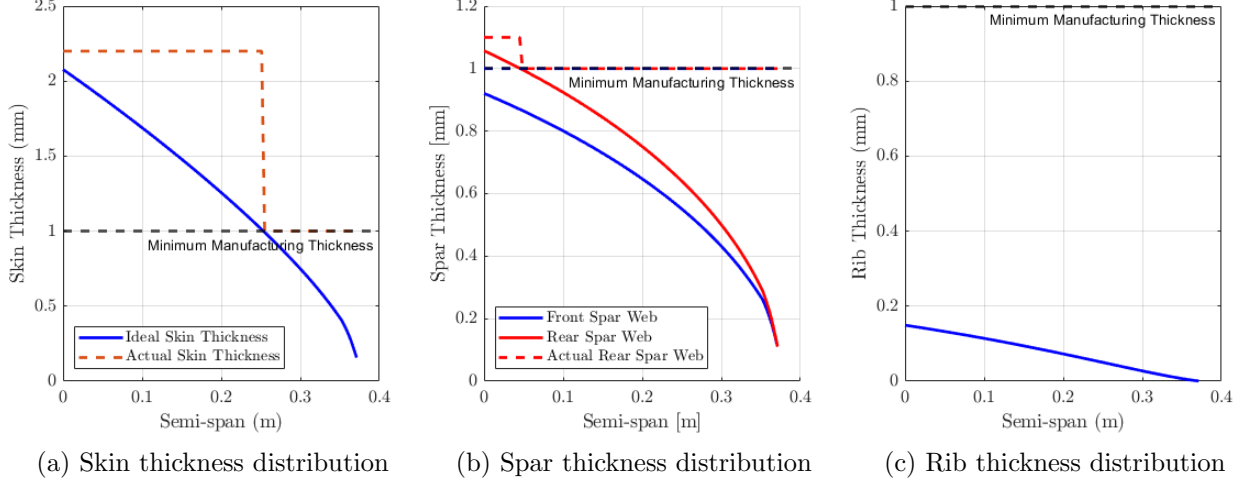


Figure 3.6: HTP Component Thicknesses

The buckling analysis of the HTP wing panels yielded a maximum combined loading ratio R_{cs} of 0.6668, far below the limit at 0.99.

3.3 Design inefficiencies & Conclusion

As figures 3.3 and 3.6 highlight, the overwhelming majority of the optimal thicknesses from the design are below the minimal manufacturing limit set as 1 mm. Since all thicknesses below that threshold will be upped to 1 mm, the structure that results is grossly over designed. Given the aim of the structure was to minimize weight for an unchanging maximum loading case, the mass increase that results from manufacturing limitations can be considered an inefficiency. Assuming that the original, optimal,

design is the minimum weight design possible, the inefficiencies of the different components of the VTP and HTP can be found in table 3.1. As the table highlights, around 38.1% of the structural mass of each VTP is inefficient, while around 40.1% of the HTP mass is inefficient. The total empenage mass inefficiency when combining those values is 39.6%. This was unfortunately to be expected for aircrafts such as the designed UAV, as its small size intrinsically implies that loads will be small. Carbon fibre would have served to make the aircraft structure much more efficient given its material properties (namely high strength and low density), but it was ruled out during material selection due to other factors, namely impact resistance and impossibility to detect cracks [1]. The final results for the weights of individual vertical and horizontal tails are in table 3.1. The total mass of the H-tail is therefore 2.485 kg.

Table 3.1: Summary of Empennage Design Results

Component	VTP				HTP			
	Skin	Spars	Ribs	Total	Skin	Spars	Ribs	Total
Optimal Mass (g)	166.4	45.0	3.2	214.6	656.7	30.3	9.7e-02	687.1
Actual Mass (g)	233.3	57.5	5.5	296.2	915.9	42.9	3.4	962.2
Inefficiency (%)	40.2	21.8	71.9	38.1	39.5	41.6	36.8	40.1
Actual Total Mass (2xVTP+2xHTP)				2485 g	Inefficiency (%)		39.6	

4 Pylon Design

After choosing VTOL as the take-off and landing method of the UAV, it was to decide where the VTOL motors would go and how they would be attached to the rest of the aircraft. Traditionally, most VTOL UAVs attach their motors to pylons emanating either from the fuselage or from the wing. After consideration, the decision was made to have the pylons emanate from the wing. This was chosen as it would constrain VTOL rotor dimensions less, while also reducing the amount of drag and air flow disturbance felt downstream of the front rotors around the root and mid-span of the wing, where the most lift is generated [18].



Figure 4.1: CAD Top and Front View of the Pylons

From this decision, the pylons were designed within a number of dimensional constraints to their position, size, cross-section, stiffness, and maximum deflection. From the various possible geometries that emerged, the pylon was then assumed to behave as a simple encastred beam problem to solve and analyse. Additional care was taken to develop a composite laminate that responded to the design criteria.

4.1 Constraints

4.1.1 Geometrical Constraints

Numerous spacial constraints were defined for the pylon location based on data and requirements from other groups.

The propulsion team defined the dimensions of the VTOL rotors radii as 30 cm from the root from their power requirements. As a result, to be able to clear the rotors from rotating over the wing, the length of the pylons from the leading edge or trailing edge of the wing was taken as the radius of the rotors with an additional safety factor of 4 cm, leading to fixed pylon lengths from the LE and TE of the wing of 0.34 m.

To allow for space to place the ailerons not too close inboard, in pink in figure 4.2, it was recommended that the pylons were at a distance superior to 0.7 m from the center-line. This position could be decided with the added requirement that the pylons be positioned and sized such that the total horizontal length from one extremity of the rotor to the other would not exceed 1.6 m, a target horizontal length for ease of use on boat decks. A short code calculated the total length at each span location, and a final length that fit this requirement was chosen as 1.54 m as figure 4.2 highlights. This length corresponds to a span location of 0.85 m, which is adequate for the aileron constraint, for a chord at that span of 0.26 m.

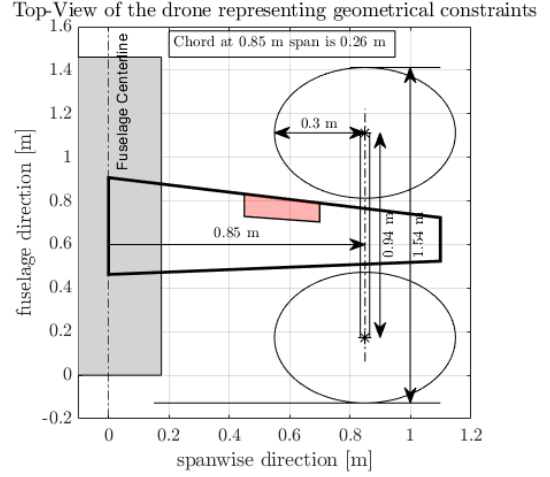


Figure 4.2: Pylon Construction Constraints

4.1.2 Cross-sectional Constraints

Constraints for the pylon cross-section were also identified. The pylons needed to be hollow to allow for the passage of wires safely from the wing and fuselage to the motor hubs for control inputs and power supply to the VTOL motors.

A hollow circular cross-section was chosen given its beneficial properties in bending but also, quite importantly, the homogeneity of its properties in all directions. This factor simplified the laminate analysis required. In combination with this constraint, the encastred beam assumption was also a limitation. For this assumption to be more accurate, the minimum wing thickness at the location of the pylon had to be superior to the outer diameter of the pylon. Having chosen a span location, the wing thickness could be obtained from [19]. It yielded a maximum height of 0.034 m. Taking a small clearance from the top and bottom surface, the maximal outer radius possible for the pylon was deemed to be $R_o = 0.016$ m, but the value selected was 0.0155 m, with the result shown in 4.3. The value of the internal radius could then be adapted for varying thicknesses to obtain the second moment of area of the section for analysis.

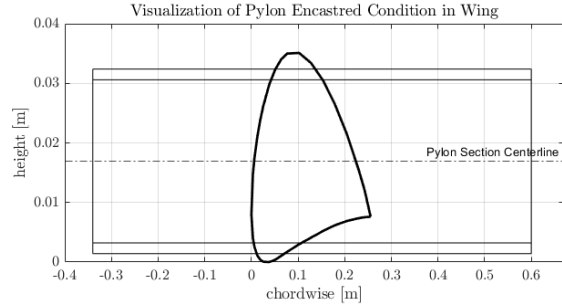


Figure 4.3: Pylon Geometry

4.2 Composite Analysis

A composite laminate was designed for the pylons. This allowed for a significant reduction in the weight of the pylons, as well as maximized strength and resistance to bending, but increased difficulty of analysis. Notable differences between designing with a regular metal and a composite laminate is that the material properties of the composite laminate will be different to those of the initial ply and that they will vary depending on the laminate stacking sequence [20]. Another important difference is the inability to currently fully predict composite failure behaviour under load, resulting in a variety of failure criterion to select and check against [21]. The analysis will focus on the maximum load case for the pylons, that is to say a single point load applied at the end of the beam due to the VTOL motor thrust upwards. The failure criteria that will be considered is if the compressive strength of

the laminate in the longitudinal direction caused by the bending moment is exceeded by the stress concentrations on the cross-section in the analysis below.

4.2.1 Laminate

The material selected for the laminate was a biaxial layup [0,90], resin infused woven fabric, of Epoxy/HS Carbon fiber [3]. This is a composite material for which there is extensive documentation, which allows for the use of carpet plots to determine some of the mechanical properties, notably compressive strength. Carpet plots are methods that rely on the distribution of the % of certain degree plies in the laminate to estimate its properties [22]. These can be found in appendix F.

For the stiffness properties to be used in the deflection analysis, the LAP Anaglyph Software [23] was used. It outputs updated values for the longitudinal and transverse stiffness moduli E_{xx} and E_{yy} as well as laminate shear stiffness G_{xy} and poisson ratio ν_{xy} , collated in table 4.2. It also outputs the different laminate stiffness matrices in the form of the ABD matrix where the A matrix is the extensional stiffness matrix, B the stretching-bending coupling matrix and D the flexural stiffness matrix. The complete report can be found in appendix G.

Table 4.1: CFRP Laminate

Layer	t (mm)	Angle (°)
1	0.15	0
2	0.15	0
3	0.15	90
4	0.15	0
5	0.15	0
6	0.15	45
7	0.15	-45
8	0.15	0
9	0.15	0
10	0.15	90
11	0.15	0
12	0.15	0
t_{total}	1.8 mm	

To decide on the laminate stacking sequence (LSS), standard industry practices were followed. These include avoiding having more than 2 plies in the same direction in a row, while also attempting to have as symmetrical of a layup as possible, keeping any non-symmetrical plies as close the centre as possible to prevent out-of-plane movements [22]. This can be seen for the $\pm 45^\circ$ plies. These were kept as non-symmetrical to produce a balanced laminate (whereby each side of the mid-plane has equivalent numbers of $\pm$ angled plies) to better resist shear and buckling. A large number of 0° plies was used to increase compressive strength, while some 90° plies were used to carry possible transverse loads [22]. This stacking resulted in the laminate mechanical properties observed in table 4.2. From the % of 0° and $\pm 45^\circ$ plies, the compressive strength was estimated to be 430 MPa and the tensile strength was estimated to be 1.1 GPa from figures F.1 and F.2. This signifies that the compressive strength of the laminate will be the limiting factor.

Table 4.2: Laminate Stiffness Properties

Membrane Stiffness (GPa)		Bending Stiffness (GPa)	plies (°)	% laminate
E_{xx}	101.54	117.76	0	66
E_{yy}	39.79	35.79	± 45	17
G_{xy}	11.13	6.22	90	17
ν_{xy}	0.23	0.13	Total	100

4.3 Analysis & Results

Having obtained the mechanical properties of the laminate, it was possible to analyze the pylons using standard ETB. The value of stiffness used in the calculations of the slope and deflections was that of the transverse bending stiffness - this value was used because it is the smallest possible value in the principal directions, and the implementation of large safety factors in composite is relevant due to their sometimes unpredictable behaviour [20].

The length of the beam was taken as the longest distance from a spar, the location where the pylons are assumed to be encastred, to the motor hub. This was for the back spar, and yielded a length L_{py} of 0.4167 m. The cross-section second moment of area is that of a hollow cylinder with a thickness

$$\sigma = \frac{\max(BM) * y}{I_{cross-section}} \quad (9)$$

of 1.8 mm. The resulting shear forces, bending moments and deflections and slopes can be visualized in figure 4.4a. The maximum slope at the end of the beam yields a 99.99% thrust vectoring for this pylon design.

Having identified that the compressive strength of the laminate was lower than its tensile strength, the stress distribution on the pylon cross section was obtained using equation 9. The maximum stress observed in the structure in compression was $\sigma_{max} = 55.21$ MPa. This value is far below the compressive strength of the composite laminate at 430 MPa, therefore the pylons are not at risk of failure under their main loading scenario. This analysis was confirmed in its accuracy by an FEA analysis performed by Rieuneau [24], which yielded similar deflection and max stress values at the connection of the skin and pylon. Other possible sources of failure like impact were not considered as they were not considered relevant for the minimal cost and mass design of the aircraft. Similarly crack resistance was not considered as the pylons are placed into the high-wing, keeping them out of the way of ground impacts and the foam in the fuselage absorbs most of the landing loads, protecting the rest of the structure including the pylons, as per Borsu Lefevres' Report [11]. The final weight for both pylons is 0.47 kg.

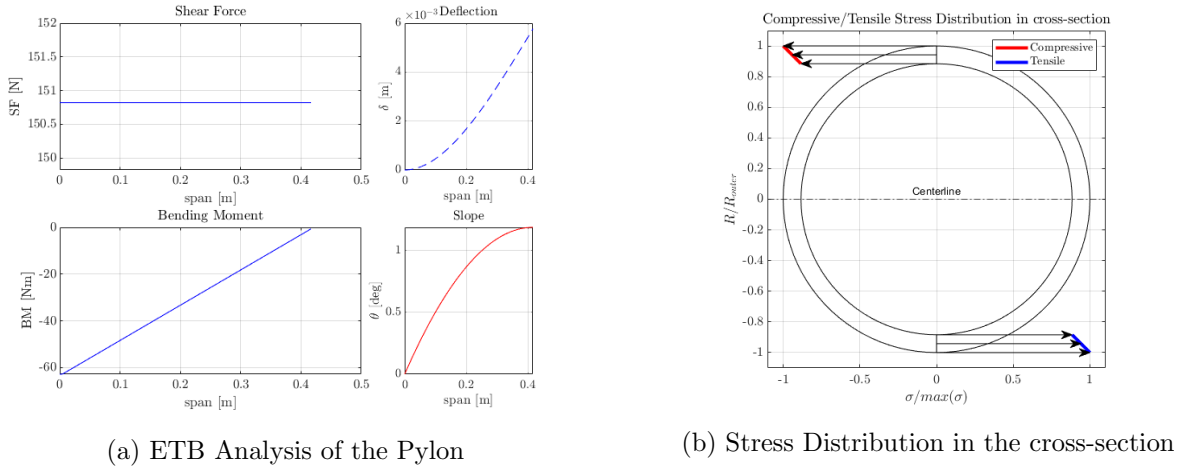


Figure 4.4: Pylon Analysis Results

5 Design Evaluation & Conclusion

The final design reached only represents a third iteration of the entire aircraft. Due to the current maximum loadings it is clear that the structural components of the empennage are overdesigned. This implies that further iterations for updated weight estimates would yield little results. This is partly because the main forces on the empennage are the lift loads, which are unlikely to change and are of much greater amplitude. But it is also because the inertia loads are likely to get smaller as a result, which means lower required thicknesses when the current manufacturing constraints of 1 mm are already limiting. Another aspect for improvement would be the replacement of the entire empennage structure by composites. This was abandoned during the material selection process due to costs, impact and crack resistance. However with proper failure analysis and lab testing of the laminates that would be designed to obtain accurate values for the mechanical properties, composites become more viable and are expected to yield weight gains of around 42% compared to aluminium [25]. The same can be said for the pylon structure. A more developed failure envelope could be obtained through lab tests. This would possibly allow for a much more efficient structure in terms of its weight-to-load capabilities. Another point for improvement of the pylon relates to their sizing. The pylons in the latest design iteration, the one used for FEA, are too large to be safely encastred in the wing. Appendix B resolves this issue for future iterations by encasing them in the spars. Nevertheless both designs are still limited in the fact that there hasn't been extensive FEA or live testing performed on them.

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A Material Selection - Additional CES Plots

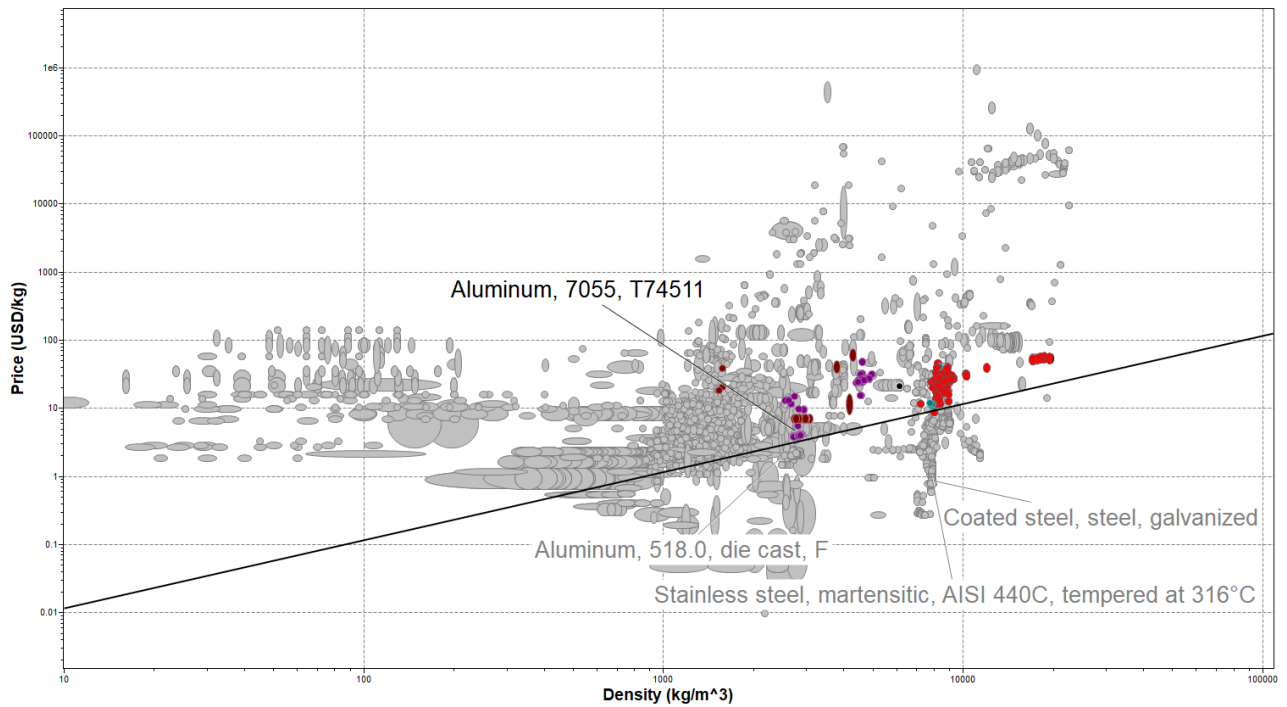


Figure A.1: Price to ρ Ashby Plot for Material Selection of load carrying structures

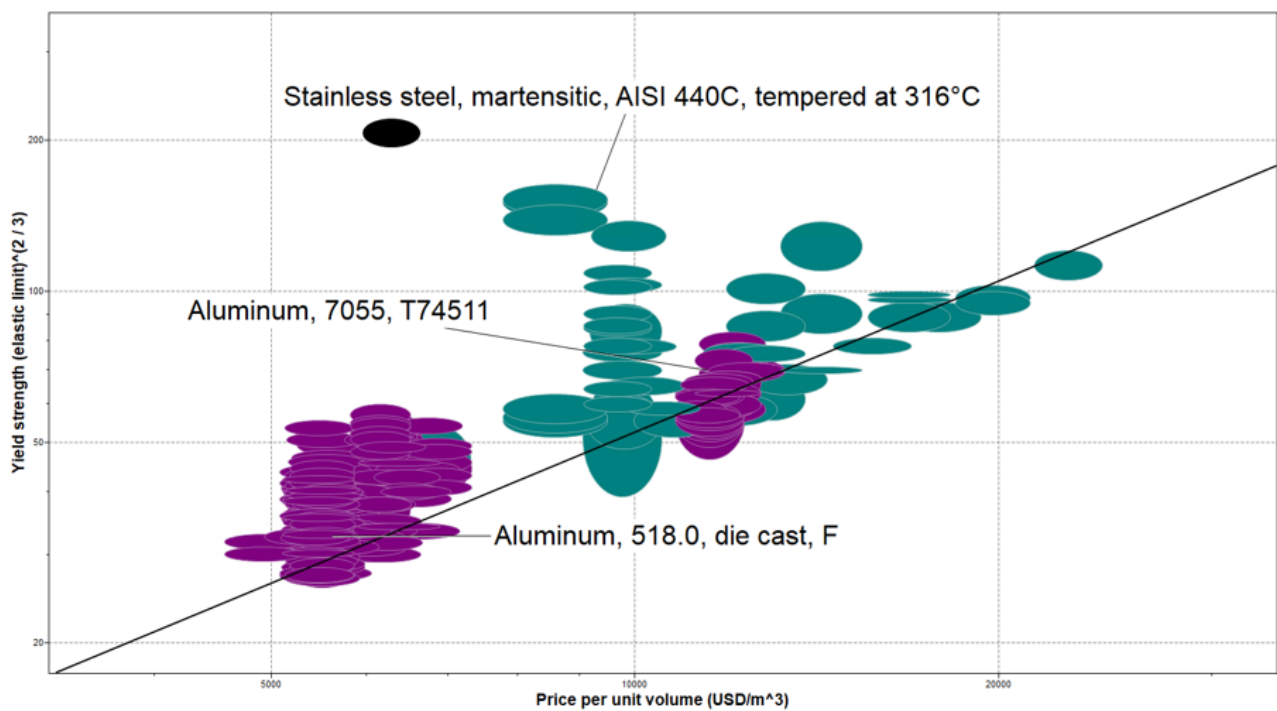


Figure A.2: Yield Strength Ashby Plot

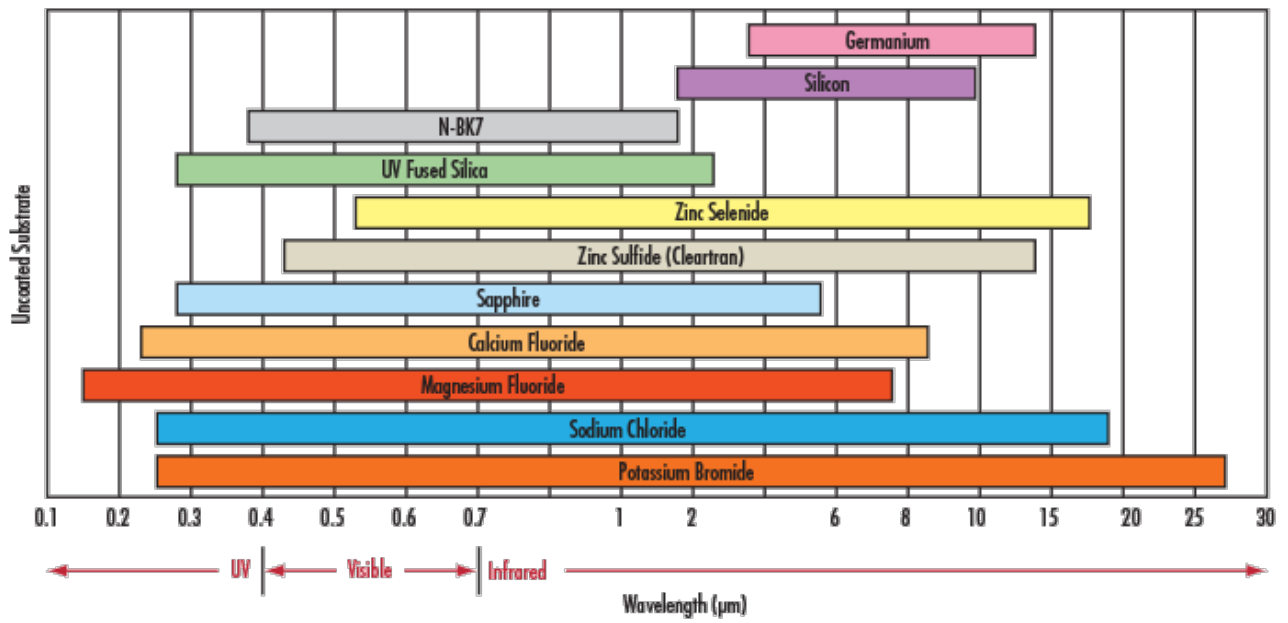


Figure A.3: LWIR Material Transparency Plot [4]

B Updated Design for the Pylon

Constraints for the pylon cross-section were also identified. The pylons needed to be hollow to allow for the passage of wires safely from the wing and fuselage to the motor hubs for control inputs and power supply to the VTOL motors. A hollow circular cross-section was chosen given its beneficial properties in bending but also, quite importantly, the homogeneity of its properties in all directions. This factor simplified the laminate analysis required. In combination with this constraint, the encastred beam assumption was also a limitation. For this assumption to be more accurate, the minimum wing spar height at the location of the pylon had to be superior to the outer diameter of the pylon. Having chosen a span location, the front and rear spar heights could be obtained from [16]. The rear spar height was the limiting factor, calculated to be $0.0789c$, or 0.0202 m. Taking a small clearance from the top and bottom surface, the maximal outer radius possible for the pylon was deemed to be $R_o = 0.0085$ m. The value of the internal radius could then be adapted for varying thicknesses to obtain the second moment of area of the section for analysis.

Having obtained the mechanical properties of the laminate, it was possible to analyze the pylons using standard ETB.

The value of stiffness used in the calculations of the slope and deflections was that of the transverse bending stiffness - this value was used because it is the smallest possible value in the principal directions, and the implementation of large safety factors in composite is relevant due to their sometimes unpredictable behaviour. The length of the beam was taken as the longest distance from a spar, the location where the pylons are assumed to be encastred, to the motor hub. This was for the back spar, and yielded a length L_{py} of 0.4167 m. The cross-section second moment of area is that of a hollow cylinder with a thickness of 1.8 mm. The resulting shear forces, bending moments and deflections and slopes can be visualized in figure B.2. The maximum slope at the end of the beam yields a 98.98% thrust vectoring.

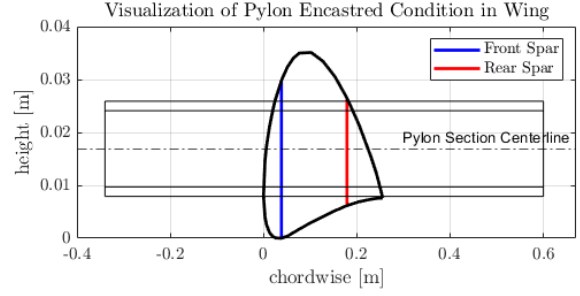


Figure B.1: Pylon Geometry

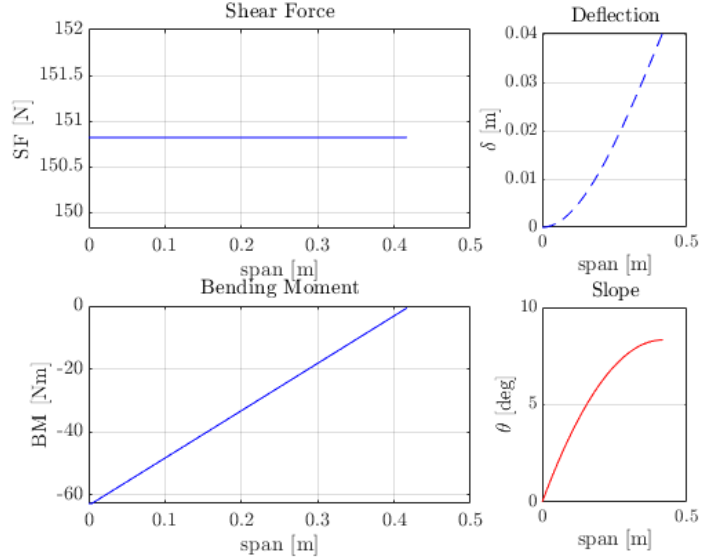


Figure B.2: ETB Results

C V-N Diagram

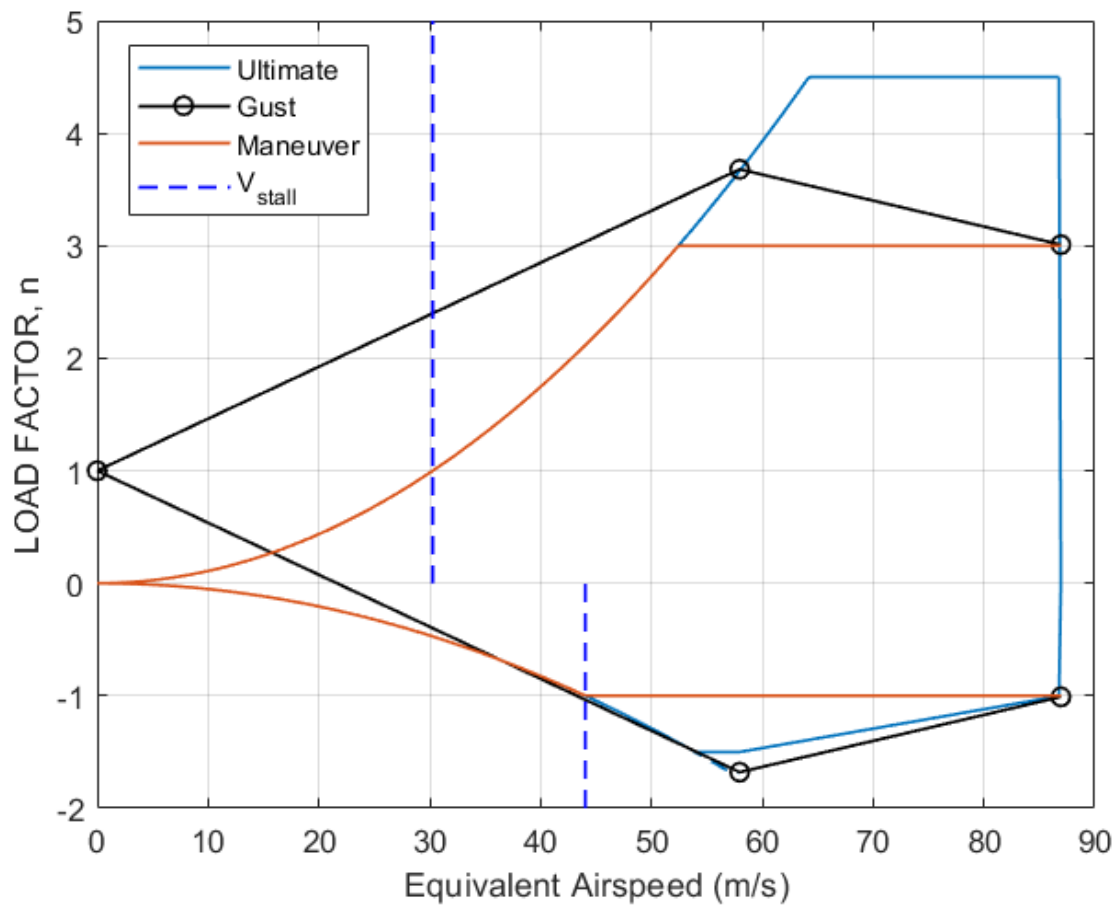


Figure C.1: V-N Diagram

D ESDU Datasheet

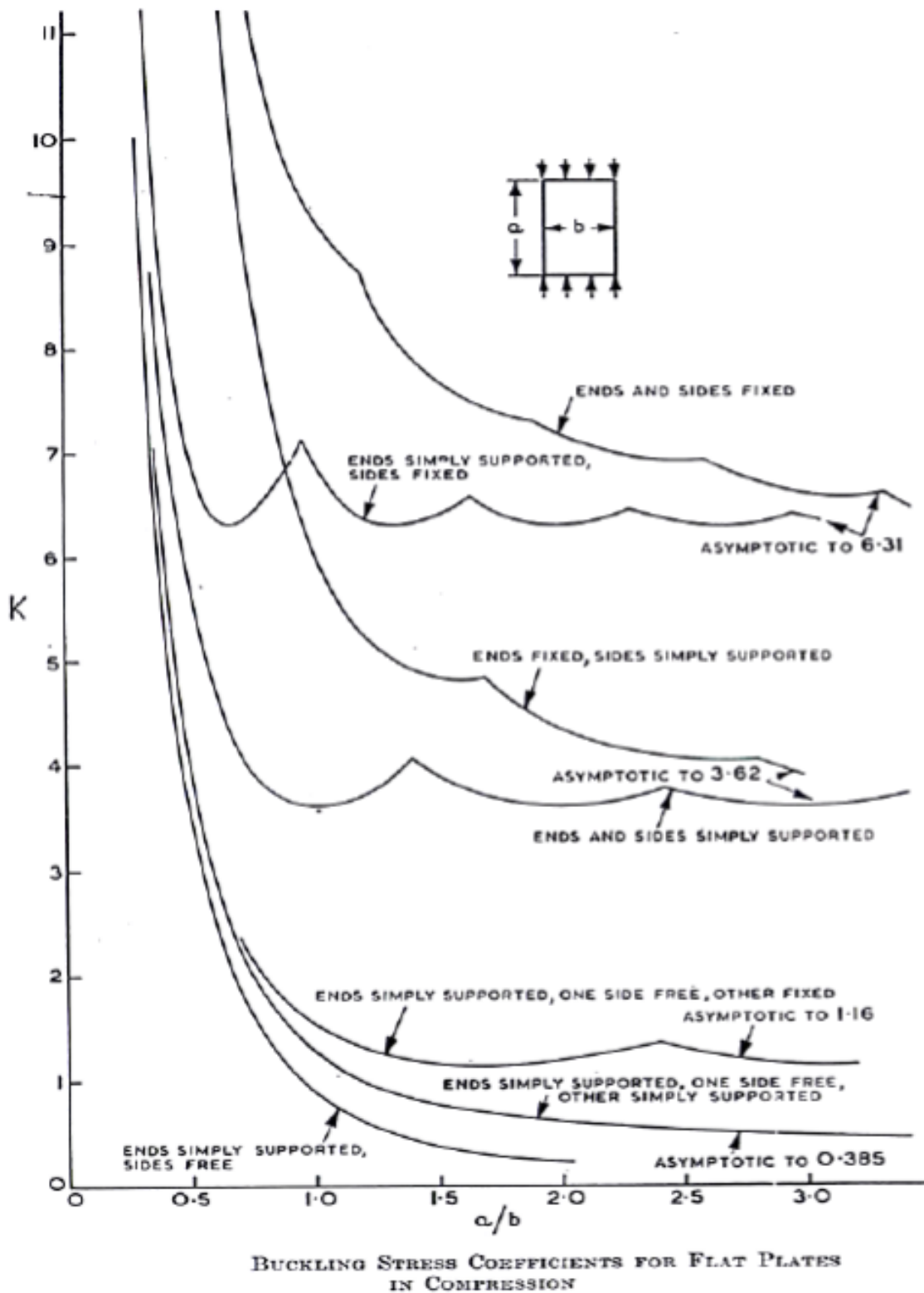


Figure D.1: ESDU Buckling Coefficient for a flat panel

E VTOL Motor Locations

From the above pylon dimensioning and positioning, the positions of the VTOL Motor locations could be determined for the control team, as seen below in table E.1.

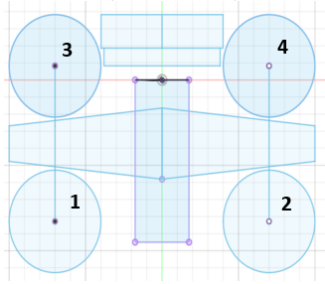


Figure E.1: Motor Locations

Table E.1: Resulting VTOL Motor Locations

VTOL Prop	from nose (m)	span (m)
1	0.16	-0.85
2	0.16	0.85
3	1.06	-0.85
4	1.06	0.85

F Carpet Plots for Epoxy/HS Carbon fiber

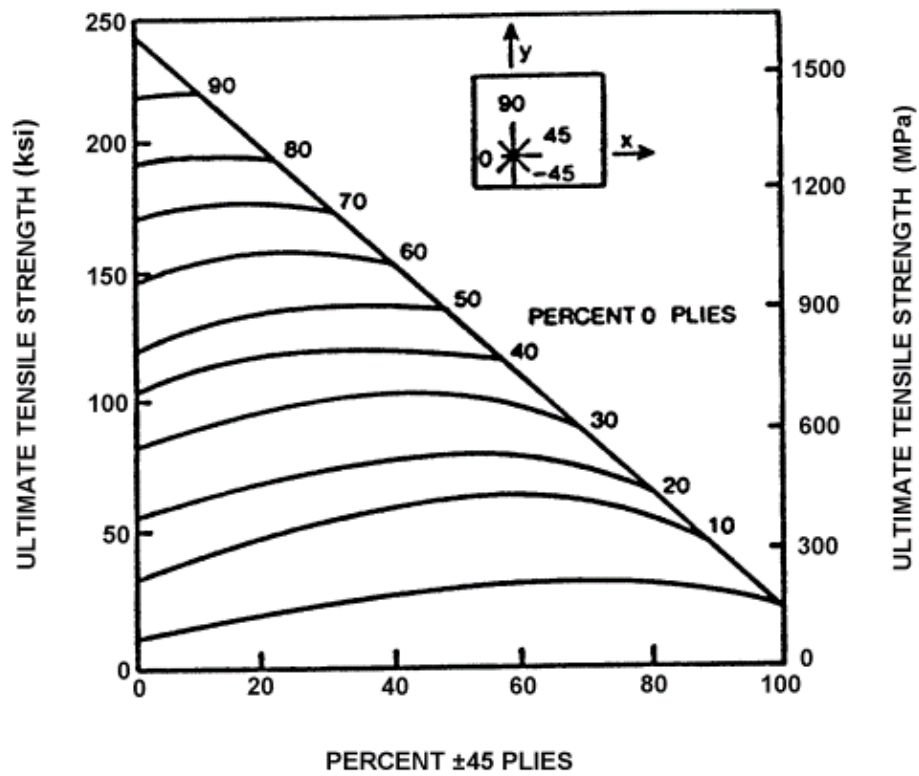


Figure F.1: Tensile Strength Carpet Plot

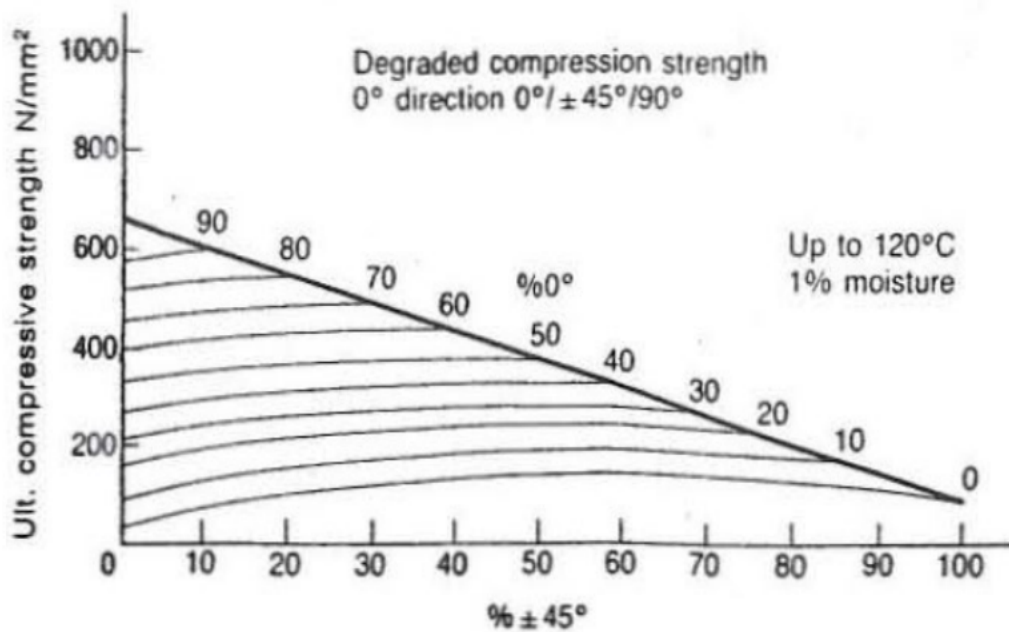


Figure F.2: Compressive Strength Carpet Plot

G Laminate LAP Software Report

The LAP Anaglyph Software Report Output can be found below.

Report LAP ANALYSIS GDP1

No description

Data File : \\icnas2.cc.ic.ac.uk\os1218\Desktop\GDP\GDPLAP.Id

Units : mm N deg °C %

Volume fraction correction method : Halpin-Tsai

Failure tracked by : Tsai-Wu Tsai-Hill Hoffman Max.stress

Selected Lay-up : Balanced CFRP

GDP Pylon LAP analysis

Lay-up is not symmetric Total number of layers = 12

Total thickness: nominal = 1.8 mm actual = 1.8 mm

Curing Temperature = 135 °C Room Temperature = 15 °C

Unnotched Compressive Strength based on Budiansky-Fleck model

KIC = (undefined)

Layer	material name	thickness (nominal, mm)	angle (deg)
--- top ---			
1 :	CF/EPOXY	0.15	0
2 :	CF/EPOXY	0.15	0
3 :	CF/EPOXY	0.15	90
4 :	CF/EPOXY	0.15	0
5 :	CF/EPOXY	0.15	0
6 :	CF/EPOXY	0.15	45
7 :	CF/EPOXY	0.15	-45
8 :	CF/EPOXY	0.15	0
9 :	CF/EPOXY	0.15	0
10 :	CF/EPOXY	0.15	90
11 :	CF/EPOXY	0.15	0
12 :	CF/EPOXY	0.15	0
-- bottom --			

Material : CF/EPOXY

Example CFRP. Source: NPL

E11	=	139000	N/mm ²
E22	=	14100	N/mm ²
G12	=	6100	N/mm ²
v12	=	0.32	
α11	=	(undefined)	
α22	=	(undefined)	
β11	=	(undefined)	
β22	=	(undefined)	
E resin	=	(undefined)	
v resin	=	(undefined)	
Vf nom.	=	0.6	
Vf act.	=	0.6	
S11T	=	1658	N/mm ²
S22T	=	46	N/mm ²
S11C	=	710	N/mm ²
S22C	=	254	N/mm ²
S12	=	78	N/mm ²
F12	=	auto	

BFS k = (undefined)
 BFS ϕ = (undefined)
 BFS β = (undefined)
 Density = 1.52e-006 kg/mm³
 Cost = 1.701e-005 \$/mm³
 Safety Factors for Strength :
 11 (T) = 1
 22 (T) = 1
 11 (C) = 1.1
 22 (C) = 1.1
 12 (S) = 1.1

Selected Loading : AXIAL + BENDING (ref: Operating Conditions)

To be used with lay-up Non-Linear CFRP

			order	steps
X Load	= 1	N/mm	1	1
Y Load	= 0	N/mm	1	1
XY Load	= 0	N/mm	1	1
X Moment	= -51	N	1	1
Y Moment	= 0	N	1	1
XY Moment	= 0	N	1	1
Top Temperature	= 50	°C		
Bottom Temperature	= -20	°C		
Moisture Content	= 1.2	% weight		
Allowable Strain and Curvature range :				
ϵ_x	= ± 1	%		
ϵ_y	= ± 1	%		
ϵ_{xy}	= ± 1	%		
κ_x	= ± 0.001	/mm		weight = 1
κ_y	= ± 0.001	/mm		weight = 1
κ_{xy}	= ± 0.001	/mm		weight = 1
Safety Factors for failure :				
11 (T)	= 1			
22 (T)	= 1			
11 (C)	= 1.1			
22 (C)	= 1.1			
12 (S)	= 1.1			

Effective Stiffness :

	Membrane	Flexural	
Exx =	101537	117755	N/mm ²
Eyy =	39786.1	35785.2	N/mm ²
Gxy =	11129.8	6222.81	N/mm ²
vxy =	0.23406	0.12923	
vyx =	0.091713	0.039271	
Elxx =		57228.8	N mm
Elyy =		17391.6	N mm
GJ =		12097.2	N mm

ABD Matrix :

		$\begin{bmatrix} \text{N/mm} & \text{N} \\ \text{N} & \text{N mm} \end{bmatrix}$			
186942	17296	0	0	0	-709.937
17296	73352.2	0	0	0	-709.937
0	0	20069.1	-709.937	-709.937	0
0	0	-709.937	57545.9	2284.02	0
0	0	-709.937	2284.02	17505.4	0
-709.937	-709.937	0	0	0	3032.77

Inverse ABD Matrix :

		$\begin{bmatrix} \text{mm/N} & 1/\text{N} \\ 1/\text{N} & 1/(\text{N mm}) \end{bmatrix}$			
5.47146e-06	-1.28064e-06	0	0	0	9.81024e-07
-1.28064e-06	1.39636e-05	0	0	0	2.96893e-06
0	0	4.9916e-05	5.38248e-07	1.95413e-06	0
0	0	5.38248e-07	1.74737e-05	-2.25805e-06	0
0	0	1.95413e-06	-2.25805e-06	5.74991e-05	0
9.81024e-07	2.96893e-06	0	0	0	0.000330656

Layer Stress graph :

Direction : XX

Components : Curing + Hygro + Thermal + Mechanical

